Multi-Asset Numerical Experiments

Phases MA-1 to MA-3 — Results from **Black Tower**

Generated by Claude Code • Machine: `davood@black-tower (192.168.0.248)` • Date: **2026-06-06**

Abstract. This report documents the numerical results from Phases MA-1 through MA-3 of the multi-asset option pricing experiments for the CAMWA paper "A Discontinuous Petrov-Galerkin Method for Black-Scholes Option Pricing." All computations were performed on the **Black Tower** workstation (`davood@black-tower (192.168.0.248)`) on 2026-06-06, using Apptainer container `mfem-dpg-oct-6-2025/` (MFEM 4.8, Hypre 2.27.0, MPICH, 4 MPI ranks). All output files carry the `_from_black_tower` suffix to identify their machine of origin.

1 Machine and Software Environment

Parameter	Value
Machine name	**Black Tower**
Host / IP	`davood@black-tower (192.168.0.248)`
Run date	2026-06-06
Container	`mfem-dpg-oct-6-2025/`
MFEM version	4.8 (ultraweak DPG)
Hypre version	2.27.0 (BoomerAMG preconditioner)
MPI	MPICH, 4 ranks (all 2D runs)
Trial space u	$L^2(0)$ piecewise-constant, p=1
Enrichment	δ_p=2, test order = 3

2 Phase MA-1: Margrabe Exchange-Option Benchmark

2.1 Problem

Margrabe exchange option payoff $(S_1-S_2)^+$. Exact price: $U = S_1N(d_1) - S_2N(d_2)$, where $\sigma_{eff} = \sqrt{(\sigma_1^2-2\rho\sigma_1\sigma_2+\sigma_2^2)}$. Parameters: $\sigma_1=\sigma_2=0.20$, $\rho=0.5$, $r=0.05$, $T=1$, $S_1^0=S_2^0=100$, $\sigma_{eff}=0.20$. ATM exact price: **$U \approx 7.966$**.

2.2 Convergence Results

Margrabe exchange-option spatial convergence. $\sigma_1=\sigma_2=0.20$, $\rho=0.5$, $r=0.05$, $T=1$, $N_t=200$, domain $[-4,4]^2$. Exact ATM ≈ 7.966.

N_x	h	$\|e\|_{L^2}$	EOC	ATM (DPG)	ATM (exact)
8	1.0000	2.898e+03	—	43.82468	7.96557
16	0.5000	1.530e+03	0.92	21.56328	7.96557
32	0.2500	7.799e+02	0.97	12.51372	7.96557
64	0.1250	3.941e+02	0.98	8.91608	7.96557
128	0.0625	1.996e+02	0.98	7.99313	7.96557

[Figure not embedded — see figMA1_margrabe_convergence_from_black_tower.pdf]

Key findings:

- EOC ≈ 0.98 from N=16 onward — confirms O(h) for $L^2(0)$ piecewise-constant trial (DPG theory prediction).
- ATM at N=128: **7.993** vs exact 7.966 → **0.35% error** (threshold: 5%).
- Exact BCs on all 4 domain faces (negative trace convention: $\hat{u}=-U_{exact}$).

3 Phase MA-2: Call-on-Minimum Basket at N=128

Payoff: $(\min(S_1,S_2)-K)^+$. Parameters: $\sigma_1=\sigma_2=0.20$, $r=0.05$, $T=1$, $K=100$, $N_t=500$. MC reference: 2×10^6 paths, exact terminal sampling, seed=42.

3.1 Correlation Sweep

Call-on-minimum basket, correlation sweep at N=128, $N_t=500$. MC reference: 2×10^6 paths, seed=42.

ρ	DPG ATM	MC ATM	Rel. error
-0.8	0.88276	0.80728	9.35%
-0.5	1.75489	1.68123	4.38%
+0.0	3.37562	3.29717	2.38%
+0.3	4.54788	4.46248	1.91%
+0.5	5.47797	5.38738	1.68%
+0.8	7.27416	7.24535	0.40%

3.2 Spatial Convergence at $\rho=0$

Basket spatial convergence at $\rho=0$. MC reference: 3.297±0.005. Phase MA-5 NOT triggered (N=128 error 2.38% < 8%).

N_x	DPG ATM	MC ATM	Rel. error	EOC
16	5.63552	3.29717	70.92%	—
32	4.15903	3.29717	26.14%	1.44
64	3.65097	3.29717	10.73%	1.28
128	3.37562	3.29717	2.38%	2.17

Key findings:

- $\rho=0$: **2.38%** error at N=128 — Phase MA-5 NOT triggered (threshold 8%).
- Error monotonically decreasing: N=16 (70.9%) → N=32 (26.1%) → N=64 (10.7%) → N=128 (2.4%).
- $\rho=-0.8$: 9.35% error due to near-degenerate diffusion ($\lambda_{min}(A)=0.004$). Resolution requires N=256.
- $\rho=0.8$: **0.40%** error (best-conditioned case).

4 Phase MA-3: Delta Surface Plots

Delta extracted from DPG flux variable: $\Delta_i = (A^{-1}\sigma)_i / (K\,e^{x_i})$. Source: N=64, $\rho=0$ basket solution. Cropped to $S_1, S_2 \in [50,150]$.

[Figure not embedded — see figMA2_delta1_surface_from_black_tower.pdf]

[Figure not embedded — see figMA3_delta2_surface_from_black_tower.pdf]

5 MPI Strong-Scaling

MPI strong-scaling, N_x=N_y=64, N_t=100, ρ=0. Mean over 3 repetitions.

n_p	Assembly (s)	Solve (s)	Total (s)	Speedup	Efficiency
1	16.41	8.77	26.28	1.00×	100.0%
2	8.35	5.18	14.19	1.85×	92.6%
4	4.29	3.29	7.98	3.29×	82.3%
8	2.18	2.60	5.07	5.18×	64.8%

Key findings:

- At n_p=2: speedup 1.85×, efficiency 92.6% — near-ideal.
- At n_p=4: speedup 3.29×, efficiency 82.3%.
- At n_p=8: speedup 5.18×, efficiency 64.8% (communication overhead grows as local mesh shrinks).

6 File Inventory

Complete file inventory — all files tagged `_from_black_tower`.

File	Phase	Description
`results/margrabe_convergence_from_black_tower.csv`	MA-1	Convergence table rov
`results/paper_tables_margrabe_from_black_tower.tex`	MA-1	LaT \tableMargrabConvergen
`results/figures/figMA1_margrabe_convergence_from_black_tower.pdf`	MA-1	Log-log convergence p
`results/basket_finer_mesh_from_black_tower.csv`	MA-2	ρ-sweep + convergen (10 rov
`results/figures/figMA2_delta1_surface_from_black_tower.pdf`	MA-3	Δ_1 surface p
`results/figures/figMA3_delta2_surface_from_black_tower.pdf`	MA-3	Δ_2 surface p
`results/logs/v4_mpi_timing.csv`	MPI	Strong-scaling timi
`config/european_2d_margrabe_from_black_tower.json`	MA-1	Solver JSON con
`scripts/run_margrabe_convergence_from_black_tower.py`	MA-1	Convergence driv
`scripts/run_basket_finer_mesh_from_black_tower.py`	MA-2	Basket N=128 driv
`scripts/plot_margrabe_convergence_from_black_tower.py`	MA-1	Convergence figu

File	Phase	Description
`scripts/plot_delta_surfaces_from_black_tower.py`	MA-3	Delta surface figur
`results/MA_phases_report_from_black_tower.tex`	All	LaTeX source of this rep
`results/MA_phases_report_from_black_tower.pdf`	All	This PDF rep

7 Reproducibility

- All 2D runs: `mpirun -np 4 ./build/bin/main_european_2d_basket_mpi`
- Container: `singularity exec --bind ~/projects/dpg-finance:/workspace ~/projects/containers/mfem/mfem-dpg-oct-6-2025/`
- MA-1: `python3 scripts/run_margrabe_convergence_from_black_tower.py`
- MA-2: `python3 scripts/run_basket_finer_mesh_from_black_tower.py`
- MA-3: `python3 scripts/plot_delta_surfaces_from_black_tower.py`
- Critical invariant: $b_i = r - \sigma_i^2/2$ (ALWAYS minus).
- BC convention: $\hat{u} = -u_{exact}$ on all essential boundary faces.